

Standardized Diagnostic Protocol: Hematology, Dentistry & Neuro-Oncology

Clinical Decision Support Guidelines • Document ID: MY-PRT-2026-V1

1. Clinical Scope & Objectives

This protocol defines the standardized algorithmic thresholds and triage escalation pathways across the three diagnostic domains of Moyassar Health AI: automated Complete Blood Count (CBC) analysis, deep-learning dental pathology detection, and axial MRI brain tumor classification.

2. Module I: Hematology & CBC Anemia Triage

The hematology engine analyzes laboratory hemogram parameters to classify anemia phenotypes according to clinical decision trees:

Anemia Subtype	Diagnostic Parameters	Clinical Differential	Triage Action
Microcytic Hypochromic	MCV < 80 fL, MCH < 27 pg	Iron Deficiency, Thalassemia	Order Ferritin, Iron Binding Capacity
Normocytic Normochromic	MCV 80-100 fL, Normal MCH	Anemia of Chronic Disease, Renal	Evaluate Creatinine, Reticulocyte Count
Macrocytic Megaloblastic	MCV > 100 fL, Elevated RDW	B12 Deficiency, Folate Deficiency	Order Serum B12 & Folate Assay

3. Module II & III: Neuro-Oncology & Dental Protocols

- **Brain MRI Protocol:** Axial T1-contrast series evaluated for lesion circumscription, dural attachment, and perilesional edema. Findings automatically synchronize with neurosurgical consultation scheduling.
- **Dental Pathology Protocol:** Detects pulp involvement, periapical rarefaction, and structural crown margins with automated referral to endodontic or restorative clinics.

4. Claude AI Multilingual Report Generation

To address health literacy disparities, every clinical diagnostic event is paired with dual reporting: a concise, evidence-based SOAP evaluation for medical professionals, and an empathetic, simplified Arabic and English guide empowering patients to understand their health trajectory.